Aayan Akram

437-518-4241 | akrama11@mcmaster.ca | linkedin.com/in/aayana/ | Portfolio

Education

McMaster University — Hamilton, ON, Canada
B.Eng, Computer Engineering — *Expected May 2029*

Experience

Research Assistant – ETF Lab — May 2026 – Present
Faculty of Engineering – McMaster University — *Hamilton, ON, Canada*

- Developed a MATLAB radar-processing and validation pipeline to reconstruct UAV trajectories from simulated detections and compare Constant-Velocity and Constant-Acceleration Kalman state-estimation models
- Built a reproducible 35-trial dataset across seven UAV manoeuvre classes, engineering 29 radar- and tracker-derived temporal motion features for UAV manoeuvre classification
- Trained and evaluated UAV manoeuvre classifiers using grouped Leave-One-Trial-Out cross-validation, achieving 80.58% accuracy and 0.806 macro-F1 with a tuned Random Forest

IoT Systems Developer — July 2026 – Present
Kaizen Radical — *Toronto, ON, Canada*

- Prototyping and debugging microcontroller-based control modules, integrating physical inputs and indicators while validating power, signal, and communication interfaces
- Developing embedded firmware to process real-time inputs, transmit device events over Wi-Fi to Odoo, and control local indicators from ERP status messages

Embedded Computer Vision Intern — April 2026 – July 2026
Confidential Early-Stage Venture — *Arlington, VA, USA*

- Developed mobile object-detection modules using Python, OpenCV, YOLO, and NVIDIA Jetson hardware, integrating vision software with physical embedded systems for real-time inference
- Prepared and trained custom datasets using Roboflow, then tested and refined model performance for deployment under edge-compute and latency constraints

Projects

Autonomous Search-and-Rescue UAV | *ArduPilot, MAVLink, Python, OpenCV, Raspberry Pi*

- Developing a 6S, 10-inch autonomous UAV integrating ArduPilot, MAVLink-based companion computing, GPS navigation, and onboard computer vision for time-sensitive search-and-rescue missions
- Engineering propulsion, power, vibration, thermal, and communication subsystems while developing modular compute and payload interfaces for swappable mission hardware

STM32-Based Flight Controller | *STM32, Altium Designer, Embedded C, LTspice, SPI, I2C, UART*

- Designing a custom STM32F405-based flight controller in Altium, including multi-stage power regulation, MCU clocking and boot circuitry, sensor and communication interfaces, motor outputs, programming access, and input protection
- Developing PCB and firmware architectures for real-time sensor acquisition, attitude estimation, and motor control, with design considerations for power integrity, EMI mitigation, hardware bring-up, and flight validation

Additional projects and technical work: aayanakram.vercel.app

Additional Experience

Accelerator Physics Intern | *STFC Daresbury Laboratory, Warrington, UK* — July 2024 – August 2024

- Supported accelerator and vacuum-system work while using Molflow and Opera2D to simulate beamline vacuum conditions and magnetic fields

Systems Engineering Intern | *Arm, Manchester, UK* — May 2024

- Programmed Arm-based microcontrollers for satellite telemetry, radio communication, and collision-warning prototypes tested across multiple teams

Simulation Engineering Intern | *Siemens, Manchester, UK* — February 2024

- Modelled and calibrated electromechanical digital twins in Siemens Simcenter and applied simulation-led design to a smart-building retrofit project

Technical Skills

Programming & Systems: C, C++, Python, MATLAB/Simulink, Linux, FreeRTOS
Embedded & Robotics: STM32, ESP32, Raspberry Pi, ArduPilot, MAVLink, ROS 2, I2C, SPI, UART
Engineering Tools: Altium Designer, LTspice, OpenCV, YOLOv8, Roboflow, SOLIDWORKS